1) Create a table EMP1 with following structure.

ID	Name	Basic	Designation	Age

Column	Description
ID	Number (2)
Name	Varchar2 (10)
Basic	Number (6,2)
Designation	Varchar 2 (10)
Age	Number (2)

```
CREATE TABLE EMP1
(
    ID INT,
    Name VARCHAR(10)
    Basic Decimal (6,2)
    Designation VARCHAR(10)
    Age INT
);
    DESC EMP1
```

Output:-

Field	Type	NULL key	Default	Extra.
ID	int	YES	NULL	
Name	varchar (10)	YES	NULL	
Basic	decimal (6,2)	YES	NULL	
Designation	Varchar (10)	YES	NULL	
Age	int	YES	NULL	

2) Change the data type of the field Basic from float to integer with required size of the EMP1 table.

```
ALTER TABLE EMP1
 MODIFY Basic Int;
 DESC EMP1;
```

Output:-

Field	Type	NULL key	Default	Extra.
ID	int	YES	NULL	
Name	Varchar(10)	YES	NULL	
basic	int	YES	NULL	
Designation	Varchar (10)	YES	NULL	
Age	int	YES	NULL	

1) Create a table EMPI with following structure.

ID	Name	Basic	Designation	Age

Column	Description
ID	Number (2)
Name	Varchar2 (10)
Basic	Number (6,2)
Designation	Varchar 2 (10)
Age	Number (2)

```
CREATE TABLE EMPI
  (
    ID INT,
    Name VARCHAR(10)
    Basic Decimal (6,2)
    Designation VARCHAR(10)
    Age INT
  );
    DESC EMPI
```

Output:-

Field	Type	NULL key	Default	Extra
ID	Int	YES	NULL	
Name	Varchar (10)	YES	NULL	
Basic	decimal (6,2)	YES	NULL	
Designation	Varchar (10)	YES	NULL	
Age	int	YES	NULL	

2) Change the data type of the field Basic from float to integer with required size of the EMPI table.

```
ALTER TABLE EMPI
  MODIFY Basic Int;
  DESC EMPI;
```

Output:-

Field	Type	NULL key	Default	Extra.
ID	int	YES	NULL	
Name	Varchar(10)	YES	NULL	
basic	int	YES	NULL	
Designation	Varchar (10)	YES	NULL	
Age	int	YES	NULL	

3) Change the field size of Name column of the EMP1 table from 10 to 15

```
ALTER TABLE EMP1
MODIFY Name VARCHAR(15);
DESC EMP1
```

Output:-

Field	Type	NULL key	Default	Extra
Id	int	YES	NULL	
Name	Varchar(15)	YES	NULL	
Basic	int	YES	NULL	
Designation	Varchar(10)	YES	NULL	
Age	int	YES	NULL	

4) Create another table EMP_trainee with the same (changed) structure. The column Id to be renamed as EMP_id in the EMP_trainee table.

```
CREATE TABLE EMP_trainee
(
  Emp_id INT,
  Name VARCHAR(15),
  Basic INT,
  Designation VARCHAR(10),
  Age INT
);
DESC EMP_trainee;
```

Output:-

Field	Type	NULL key	Default	Extra
Emp_id	int	YES	NULL	
Name	Varchar(15)	YES	NULL	
Basic	int	YES	NULL	
Designation	Varchar(15)	YES	NULL	
Age	int	YES	NULL	

5) Insert following data in EMP1 table:-

(1, Rohit, 6700, Manager, 24)
(2, Sunil, 6200, Engineer, 27)
(3, Payal, 6300, Engineer, 25)
(4, Kunal, 6700, Trainer, 28
(5, Sunita, 6230, Trainer, 26)
(6, Bimal, 7000, Trainer, 25)

3) Change the field size of Name column of the EMP1 table from 10 to 15

```
ALTER TABLE EMP1
MODIFY Name VARCHAR(15);
DESC EMP1
```

Output:-

Field	Type	NULL	key	Default	Extra
ID	int	YES		NULL	
Name	Varchar(15)	YES		NULL	
Basic	int	YES		NULL	
Designation	Varchar(10)	YES		NULL	
Age	int	YES		NULL	

4) Create another table EMP_trainee with the same (changed) structure. The column ID to be renamed as EMP_id in the EMP_trainee table.

```
CREATE TABLE EMP_trainee
(
  Emp_id INT;
  Name VARCHAR(15),
  Basic INT,
  Designation VARCHAR(10),
  Age INT
);
DESC EMP_trainee;
```

Output:-

Field	Type	NULL	key	Default	Extra
Emp_id	int	YES		NULL	
Name	Varchar(15)	YES		NULL	
Basic	int	YES		NULL	
Designation	Varchar(15)	YES		NULL	
Age	int	YES		NULL	

5) Insert following data in EMP1 table:-

(1, Rohit, 6700, Manager, 24)
(2, Sunit, 6200, Engineer, 27)
(3, Payal, 6300, Engineer, 25)
(4, Kunal, 6700, Trainer, 28
(5, Sunita, 6250, Trainer, 26)
(6, Bimal, 7000, Trainer, 25)

(1, 'Rohit', 6700, 'Manager', 24)
INSERT INTO EMPI VALUES
(2, 'Sunil', 6200, 'Engineer', 27)
INSERT INTO EMPI VALUES
(3, 'Payal', 6300, 'Engineer', 25)
INSERT INTO EMPI VALUES
(4, 'Kunal', 6700, 'Trainee', 28)
INSERT INTO EMPI VALUES
(5, 'Sunita', 6230, 'Trainee', 26)
INSERT INTO EMPI VALUES
(6, 'Bimal', 7000, 'Trainee', 25)
SELECT * FROM EMPI;

Output-

ID	Name	Basic	Designation	Age
1	Rohit	6700	Manager	24
2	Sunil	6200	Engineer	27
3	Payal	6300	Engineer	25
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

6) Insert all rows with the designation 'trainee' from the EMPI table to EMP-trainee table.

```
INSERT INTO EMP_trainee
SELECT ID, Name, Basic, Designation, Age
FROM EMPI
WHERE Designation = 'Trainee';
SELECT * FROM EMP_trainee;
```

Output-

EMP_id	Name	Basic	Designation	Age
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

7) Add columns Skills (data type-varchar 2 & size=10) & DOJ (data type-date) to the EMPI table & add data for the skills & DOJ columns according to your own wish.

```
ALTER TABLE EMPI
ADD Skills VARCHAR(10)
ADD DOJ DATE;
UPDATE EMPI
SET Skills = 'Python', DOJ = '2024-07-01' WHERE ID = 1;
UPDATE EMPI
SET Skills = 'Java', DOJ = '2024-07-05' WHERE ID = 2;
UPDATE EMPI
SET Skills = 'C++', DOJ = '2024-07-10' WHERE ID = 3;
```

```
INSERT INTO EMPI VALUES
(1, 'Rohit', 6700, 'Manager', 24)
INSERT INTO EMPI VALUES
(2, 'Sunil', 6200, 'Engineer', 27)
INSERT INTO EMPI VALUES
(3, 'Payal', 6300, 'Engineer', 25)
INSERT INTO EMPI VALUES
(4, 'Kunal', 6700, 'Trainee', 28)
INSERT INTO EMPI VALUES
(5, 'Sunita', 6230, 'Trainee', 26)
INSERT INTO EMPI VALUES
(6, 'Bimal', 7000, 'Trainee', 25)
SELECT * FROM EMPI;
```

Output -

ID	Name	Basic	Designation	Age
1	Rohit	6700	Manager	24
2	Sunil	6200	Engineer	27
3	Payal	6300	Engineer	25
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

6) Insert all rows with the designation 'trainee' from the EMPI table to EMP-trainee table.

```
INSERT INTO EMP_trainee
SELECT ID, Name, Basic, Designation, Age
FROM EMPI
WHERE Designation = 'Trainee';
SELECT * FROM EMP_trainee;
```

Output -

EMP_id	Name	Basic	Designation	Age
4	Kunal	6700	Trainee	28
5	Sunita	6230	Trainee	26
6	Bimal	7000	Trainee	25

7) Add columns Skills (data type-varchar 2 & size = 10) & DOJ (data type-data) to the EMPI table & add data for the skills & DOJ columns according to your own wish.

```
ALTER TABLE EMPI
ADD Skills VARCHAR(10)
ADD DOJ DATE;
UPDATE EMPI
SET Skills = 'Python', DOJ = '2024-07-01' WHERE ID = 1;
UPDATE EMPI
SET Skills = 'Java', DOJ = '2024-07-05' WHERE ID = 2;
UPDATE EMPI
SET Skills = 'C++', DOJ = '2024-07-10' WHERE ID = 3;
```

```
UPDATE EMPI
SET Skills = 'Java', DOJ = '2024-07-15' WHERE ID = 4;
UPDATE EMPI
SET Skills = 'SQL' , DOJ= '2024-07-20' WHERE ID = 5;
UPDATE EMPI
SET Skills = 'HTML', DOJ = '2024-07-25' WHERE ID = 6;
SELECT*FROM EMPI
```

Output -

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	6700	Trainee	28	Java	2024-07-15
5	Sunita	6230	Trainee	26	SQL	2024-07-20
6	Bimal	7000	Trainee	25	HTML	2024-07-25

8) Update more than one row in one query in EMPI table

```
UPDATE EMPI
SET Basic = Basic + 500 WHERE Designation = 'Trainee';
SELECT * FROM EMPI
```

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

9) Rename the column Age of EMPI table to Age_in_years.

```
ALTER TABLE EMPI
RENAME COLUMN Age TO
Age_in_years;
DESC EMPI
```

Output -

Field	Type
ID	int
Name	varchar(15)
Basic	int
Designation	varchar(10)
Age_in_years	int
Skills	varchar(10)
DOJ	data

```
SELECT * FROM EMPI;
```

```
UPDATE EMPI
SET Skills = 'Java', DOJ = '2024-07-15' WHERE ID = 4;
UPDATE EMPI
SET Skills = 'SQL' , DOJ = '2024-07-20' WHERE ID = 5;
UPDATE EMPI
SET Skills = 'HTML', DOJ = '2024-07-25' WHERE ID = 6;
SELECT * FROM EMPI
```

Output -

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	6700	Trainee	28	Java	2024-07-15
5	Sunita	6230	Trainee	26	SQL	2024-07-20
6	Bimal	7000	Trainee	25	HTML	2024-07-25

8) Update more than one row in one query in EMPI table

```
UPDATE EMPI
SET Basic = Basic + 500 WHERE Designation = 'Trainee';
SELECT * FROM EMPI
```

ID	Name	Basic	Designation	Age	Skills	DOJ
1	Rohit	6700	manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

9) Rename the column Age of EMPI table to Age_in_years.

```
ALTER TABLE EMPI
RENAME COLUMN Age TO
Age_in_years;
DESC EMPI
```

Output -

Field	Type
ID	int
Name	varchar(15)
Basic	int
Designation	varchar(20)
Age_in_years	int
Skills	varchar(10)
DOJ	data

```
SELECT * FROM EMPI;
```

Output -

ID	Name	Basic	Designation	Age_in_years	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

10) Drop the Age column from the EMP_trainee table.

```
ALTER TABLE EMP_trainee
DROP COLUMN Age;
DESC EMP_trainee;
```

Output -

Field	Type
EMP_id	int
Name	Varchar(15)
Basic	int
Designation	Varchar(10)

Display Data -

```
SELECT * FROM EMP_trainee;
```

Emp_id	Name	Basic	Designation
4	Kunal	6700	Trainee
5	Sunita	6230	Trainee
6	Bimal	7000	Trainee

11) Rename the table EMP to EMP_Mgr_Engr

```
RENAME TABLE EMP1 to EMP_Mgr_Engr;
SHOW TABLES;
```

Output -

Tables_in_dbms
EMP_Mgr_Engr
EMP_trainee

Display Table -

```
SELECT * FROM EMP_Mgr_Engr;
```

Output -

ID	Name	Basic	Designation	Age_in_years	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

Output -

ID	Name	Basic	Designation	Age_in_years	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

10) Drop the Age column from the EMP_trainee table.

```
ALTER TABLE EMP_trainee
DROP COLUMN Age;
DESC EMP_trainee;
```

Output -

Field	Type
EMP_id	int
Name	Varchar(15)
Basic	int
Designation	Varchar(10)

Display Data -

```
SELECT *FROM EMP_trainee;
```

Emp_id	Name	Basic	Designation
4	kunal	6700	Trainee
5	Sunita	6230	Trainee
6	Bimal	7000	Trainee

11) Rename the table EMP to EMP_Mgr_Engr

```
RENAME TABLE EMP1 to EMP_Mgr_Engr;
SHOW TABLES;
```

Output -

Tables_in_dbms
EMP_Mgr_Engr
EMP_trainee

Display Table -

```
SELECT * FROM EMP_Mgr_Engr;
```

Output -

ID	Name	Basic	Designation	Age_in_years	Skills	DOJ
1	Rohit	6700	Manager	24	Python	2024-07-01
2	Sunil	6200	Engineer	27	Java	2024-07-05
3	Payal	6300	Engineer	25	C++	2024-07-10
4	Kunal	7200	Trainee	28	Java	2024-07-15
5	Sunita	6730	Trainee	26	SQL	2024-07-20
6	Bimal	7500	Trainee	25	HTML	2024-07-25

1 Name column of the EMP1

12) Truncate EMP_Mgr_Engr table.

```
TRUNCATE TABLE EMP_Mgr_Engr
SELECT * FROM EMP_Mgr_Engr;
```

Check Table structure -

```
DESC EMP_Mgr_Engr;
```

Output -

Field	Type
ID	int
Name	varchar(15)
Basic	int
Designation	varchar(20)
Age_in_years	int
Skills	varchar(20)
DOJ	date

11/09/26

2